GCE A Level H2 Computing Study Guide

Module C3: Computer Networking (Syllabus 9569)

Syllabus Overview & Module Scope
This study guide provides a comprehensive, rigorous overview of Module C3 (Computer Networking) for the GCE A Level H2 Computing 9569 Syllabus. It covers network fundamentals, physical topologies, hardware, IP addressing, transport layer protocols, network infrastructure services, the 7-layer OSI Reference Model, and an exhaustive glossary.

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Section 1: Computer Network Fundamentals & Architectures

1.1 Basic Principles & Network Classifications

A computer network is defined as a system of interconnected autonomous host devices and intermediary nodes that exchange data over shared transmission media using standardized communication protocols.

- **Host:** An endpoint host computer or server that originates or receives data streams (e.g., workstations, web servers, file servers, network printers).
- **Node:** Any active physical or logical connection point attached to a network, including Network Interface Cards (NICs), switches, and routers.

Networks are categorized primarily by their geographical coverage and administrative scale:

- **Personal Area Network (PAN):** Connects personal devices (smartphones, tablets, wearables) within an individual operating space (~10 meters) via Bluetooth, Infrared, NFC, or USB cables.
- **Local Area Network (LAN):** Interconnects hosts and peripherals within a localized environment such as a room, office, or building. Usually privately owned and managed.
- **Wireless Local Area Network (WLAN):** A LAN that employs wireless access points (APs) or Wi-Fi routers using radio frequency signals instead of physical copper cables.
- **Campus Area Network (CAN):** Interconnects multiple distinct LANs within a defined geographic property, such as a university campus or corporate complex.
- **Metropolitan Area Network (MAN):** Spans an entire city or town, typically utilizing high-speed optical fiber backbones to link distributed corporate LANs.
- **Storage Area Network (SAN):** A high-speed, dedicated network isolated from general LAN traffic that links servers directly to enterprise block-level storage disk arrays.

- **Wide Area Network (WAN):** Encompasses expansive geographical zones across regions, countries, or globally. The global Internet is the largest example of a WAN.

1.2 Network Operational Models

Network operational architectures determine how tasks, file storage, and security management are distributed among connected nodes.

Peer-to-Peer (P2P) Architecture

In a Peer-to-Peer network, individual workstations connect directly to share files, printers, and bandwidth without a centralized controlling server. Every node operates simultaneously as both a client and a server.

- **Advantages:** Inexpensive to deploy; requires no dedicated server hardware or specialized network operating system (NOS); straightforward configuration.
- **Disadvantages:** Lack of centralized security policy, backup routines, or file access auditing; network performance degrades on individual nodes when external peers access local shared folders.

Client-Server Architecture

Client-Server networks rely on dedicated, high-performance central computers (servers) that respond to requests initiated by subordinate client workstations.

- **Advantages:** Centralized management of user authentication, security permissions, file storage, and automated backups; highly scalable and capable of supporting thousands of concurrent users.
- **Disadvantages:** High initial cost for enterprise server hardware and NOS licenses; requires dedicated network administrators; central server failure disrupts connected client operations.

Section 2: Physical Layer & Network Topologies

2.1 Physical vs. Logical Topologies

Network topology defines the arrangement of nodes and links within a communications network:

- **Physical Topology:** The physical, visual layout of cabling, media, and hardware devices in space.
- **Logical Topology:** The electronic path that data signals take through the transmission media, regardless of physical cable routing.

2.2 Core Topology Configurations

- **Point-to-Point (P2P):** Direct physical link between exactly two network nodes; forms the baseline building block for point-to-point leased lines.
- **Point-to-Multipoint (P2M):** A central base station broadcasts to multiple subscriber nodes; widely used in wireless cellular and satellite networks.
- **Bus Topology:** All nodes attach directly to a single shared central trunk cable (the bus). Signals broadcast along the cable; attached nodes inspect destination addresses and ignore irrelevant

frames. Both ends of the trunk cable must be terminated with electrical resistors (terminators) to absorb signals and prevent reflections (signal bounce) that would collapse the network.

• **Ring Topology:** Nodes connect sequentially in a closed loop without exposed ends, eliminating the need for terminators. Signals travel unidirectionally from node to node, with each node repeating the signal forward. A single cable break or failed node disrupts the entire ring.

• **Star Topology:** Every host connects via a dedicated cable directly to a central intermediary device (a Layer 2 switch). Highly fault-tolerant (a broken client cable affects only that node), scalable, and easy to troubleshoot. Failure of the central switch crashes the entire segment. This is the standard layout for modern wired Ethernet LANs.

• **Mesh Topology:** In a Fully Meshed topology, every node possesses a direct physical link to every other node. Total connections required: $C = n(n - 1) / 2$ (where n is the number of nodes). In a Partially Meshed topology, critical nodes have redundant links while secondary nodes connect to only one or two devices. Provides maximum redundancy, but is cabling-impractical for large wired networks.

• **Hybrid Topologies:** Combines two or more distinct physical or logical topologies. A Star-Bus topology (physical star cabling running into a central switch acting logically as a bus) represents the most widespread Ethernet implementation.

Section 3: Network Hardware & Intermediary Devices

3.1 Transmission Media

• **Copper Cabling:** Twisted pair (Ethernet Cat 5e/Cat 6) and coaxial cables; transmits binary data via electrical voltage pulses.

• **Fiber Optic Cabling:** Transmits data as pulses of light through high-purity glass strands. Offers high bandwidth, negligible attenuation over long distances, and immunity to electromagnetic interference (EMI).

• **Undersea Optical Cables:** Thousands of miles of fiber optic cables laid across ocean floors form the physical backbone of global internet data transmission.

3.2 End-User Interfaces & Identifiers

• **Network Interface Card (NIC):** Integrated hardware controller or expansion card that enables a computer host to connect to a physical or wireless network.

• **Media Access Control (MAC) Address:** A globally unique 48-bit (6-octet) hexadecimal hardware address hardcoded into the NIC firmware at the factory. Consists of an Organizational Unique Identifier (OUI) and a device serial number, providing over 280 trillion unique address combinations.

3.3 Intermediary Devices: Hubs, Switches, & Routers

Intermediary network devices direct data frames and packets within and between network segments:

Device	OSI Layer	Intelligence & Forwarding Mechanism	Traffic & Security Impact
Hub	Layer 1 (Physical)	Unintelligent: Multi-port repeater. Blindly rebroadcasts incoming data to all connected ports.	High network congestion, severe packet collisions, high risk of eavesdropping.
Switch	Layer 2 (Data Link)	Intelligent: Inspects destination MAC addresses and forwards traffic unicast using its SAT.	Isolates traffic conversations, grants full bandwidth per port, minimizes collision domains.
Router	Layer 3 (Network)	Gateway: Examines IP addresses and routes packets between distinct networks using routing tables.	Filters packets by IP, blocks local network broadcasts from crossing external boundaries.

3.4 Switch Operation: Source Address Table (SAT)

A Layer 2 switch dynamically constructs and updates a Source Address Table (SAT) to optimize local frame forwarding:

- **Learning Phase:** When powered on, the SAT is empty. When host A transmits a frame, the switch inspects the Source MAC Address and associates host A's MAC with the physical port on which it arrived.
- **Forwarding Phase:** If the Destination MAC Address is recorded in the SAT, the switch delivers the frame exclusively to that target port (unicast forwarding). If the Destination MAC is unknown, the switch floods (broadcasts) the frame out of all ports except the incoming port. Once the target host replies, its MAC and port are learned and stored.

Section 4: Network Addressing & Subnetting

4.1 MAC Address vs. IP Address Comparison

- **MAC Address (Layer 2):** Permanent 48-bit physical identifier hardcoded into hardware. Defines host identity ('who you are') within a local switched network.
- **IP Address (Layer 3):** Dynamic 32-bit (IPv4) or 128-bit (IPv6) logical address assigned via software/DHCP. Defines network location ('where you are') across global routed networks.
- **Dual Addressing Requirement:** MAC addresses enable local switches to deliver data directly between neighboring NICs; IP addresses enable global routers to direct packets across intermediate networks.

4.2 IP Address Formats: IPv4 vs. IPv6

- **IPv4:** 32-bit address represented in dotted-decimal notation (4 octets, e.g., 192.168.0.10). Yields ~4.3 billion unique addresses, which are now exhausted globally.
- **IPv6:** 128-bit address written in hexadecimal separated by colons (e.g., 2001:0db8:85a3::8a2e:0370:7334). Provides 3.4 x 10^38 unique addresses.

4.3 Subnet Masks & Local vs. Remote Routing Decisions

A Subnet Mask is a 32-bit binary number composed of contiguous 1s (representing the Network ID) followed by contiguous 0s (representing the Host ID). For example, 255.255.255.0 equals 24 contiguous 1s (shorthand notation /24).

Before transmitting data, a sending host performs a bitwise logical AND operation between its own IP address / subnet mask and the destination IP address:

- **Local Delivery:** If the calculated Network IDs match, the destination is on the same local network segment. The host uses the Address Resolution Protocol (ARP) to resolve the destination IP into a MAC address and transmits the frame directly via the local switch.
- **Remote Delivery:** If the calculated Network IDs differ, the destination host resides on an external network. The host encapsulates the packet and forwards it to its configured Default Gateway (the local router interface) for routing.

Section 5: Transport Layer Protocols & Data Switching

5.1 Circuit Switching vs. Packet Switching

- **Circuit Switching:** Establishes a dedicated, reserved physical channel between sender and receiver for the entire duration of communication (e.g., traditional landline phone networks). Guarantees zero jitter and fixed bandwidth, but wastes line capacity during periods of silence and is expensive to maintain.
- **Packet Switching:** Breaks data messages down into small, standardized chunks called packets. Packets travel independently across dynamic paths through intermediary routers, balancing network load and reassembling at the destination. Highly resilient to node failures and cost-effective; forms the core operational model of the Internet.

5.2 Structure of a Network Packet

- **Header:** Contains control instructions including source IP address, destination IP address, packet sequence number, protocol type, total packet length, and synchronization bits.
- **Payload (Body):** The actual chunk of user data being delivered.
- **Trailer (Footer):** Contains end-of-packet flag bits and error-checking mathematical values, most commonly a Cyclic Redundancy Check (CRC). The receiving node calculates its own CRC sum on the payload and compares it to the trailer value; mismatched values indicate data corruption during transit and trigger retransmission.

5.3 Transmission Control Protocol (TCP)

TCP is a connection-oriented, reliable transport layer protocol that guarantees ordered, error-checked byte stream delivery between applications.

TCP Three-Way Handshake (Connection Establishment)

Before transmitting payload data, TCP establishes a synchronized session between client and server:

- **1. SYN:** Client performs an active open by sending a TCP segment with the SYN (Synchronize) flag set and a randomized initial sequence number x.
- **2. SYN-ACK:** Server performs a passive open, acknowledging receipt by sending a segment with SYN and ACK flags set, acknowledgment number x + 1, and its own random sequence number y.
- **3. ACK:** Client returns a final ACK segment with sequence number x + 1 and acknowledgment number y + 1. Both hosts enter the ESTABLISHED state.

TCP Four-Way Handshake (Connection Termination)

TCP connections are full-duplex, requiring each direction to close independently:

- **Step 1:** Active closing party transmits a FIN (Finish) segment.
- **Step 2:** Passive receiving party returns an ACK segment.
- **Step 3:** Passive party finishes sending pending data and transmits its own FIN segment.
- **Step 4:** Active party returns a final ACK segment, entering TIME_WAIT to ensure all delayed segments clear before closing completely.

Section 6: Network Infrastructure Services & Application Protocols

6.1 Server Hardware Standards

Enterprise-grade servers operating 24/7 require specialized hardware designed for high reliability and continuous uptime:

- **Processors:** Intel Xeon or AMD EPYC processors supporting multi-socket motherboards, high core counts, and expanded cache sizes.
- **Memory:** Error-Correcting Code (ECC) RAM that detects and corrects single-bit memory corruptions in real time to prevent operating system crashes.
- **Storage:** Hot-swappable enterprise hard drives arranged in Redundant Array of Independent Disks (RAID) configurations to prevent data loss upon drive failure.
- **Power & OS:** Dual redundant power supplies connected to separate power circuits running non-stop server operating systems (Linux, Windows Server) capable of handling thousands of concurrent client connections.

6.2 Domain Name System (DNS)

DNS translates human-friendly domain names (e.g., www.yijc.edu.sg) into numerical IP addresses required by routers:

- **1. Local & Resolver Query:** Client OS checks local DNS cache. If absent, it queries the ISP Resolver Server.

- **2. Root & TLD Lookup:** Resolver queries Root Name Servers, which direct it to TLD Servers (.com, .edu, .sg).
- **3. Authoritative Resolution:** TLD server directs resolver to the domain's Authoritative Name Server, which holds the canonical A Record.
- **4. Response & Caching:** Resolver receives IP, caches it locally for quick future retrieval, and returns it to the client browser.

6.3 Dynamic Host Configuration Protocol (DHCP)

DHCP automates the network configuration of joining devices:

- **Scope:** The defined range of valid IP addresses configured by a network administrator for distribution.
- **Lease Mechanism:** IP addresses are assigned as temporary leases. Devices must periodically send lease renewal requests to retain their address; expired leases return to the scope pool.
- **DHCP Reservation:** Permanently binds a specific IP address to a critical device (e.g., network printer or server) based on its hardcoded MAC address.

6.4 Email Protocols Comparison

- **SMTP:** Simple Mail Transfer Protocol. Push-based protocol used for sending email from clients to mail servers and transferring email between servers over TCP port 25.
- **POP3:** Post Office Protocol 3. Downloads inbox messages from a mail server to a single local device and deletes the server copy by default, preventing folder synchronization across multiple devices.
- **IMAP:** Internet Message Access Protocol. Retains master copies of emails on the central server and continuously synchronizes all folders (Inbox, Sent, Drafts, custom folders) in real time across all logged-in devices.

Section 7: The OSI Reference Model

The Open Systems Interconnection (OSI) Model is a 7-layer conceptual framework developed by ISO that standardizes how communication functions are organized across network hardware and software.

Layer	Layer Name	Technical Function & Protocol Examples	Child-Friendly Analogy
Layer 7	Application	User-facing services and software network interfaces (HTTP, HTTPS, SMTP, IMAP, DNS, DHCP).	The Toy Store Window: The colorful app screen you interact with directly on your tablet.
Layer 6	Presentation	Data formatting, translation, compression, and encryption/decryption (SSL/TLS, ASCII, JPEG).	The Secret Translator: Unwraps gift boxes and turns alien code into pictures you can read.
Layer 5	Session	Establishes, manages, synchronizes, and closes dialogue sessions between software processes.	The Walkie-Talkie Rule: Saying 'Hello!', taking turns talking, and saying 'Over & Out!'
Layer 4	Transport	End-to-end reliability, segment sequencing, flow control, and error checking (TCP, UDP).	The LEGO Box Inspector: Ensures all 10 numbered blocks arrive safely and in order.
Layer 3	Network	Logical IP addressing and packet routing across separate global networks (IPv4, IPv6, ICMP).	The City Map & GPS: Your home mailing address and postal code for delivery drivers.
Layer 2	Data Link	Physical node-to-node frame delivery across switches using MAC addresses and SAT tables.	The Local Name Badge: Checks hardware name tags so neighbors don't get your mail.
Layer 1	Physical	Raw bitstream transmission over copper cables, fiber optics, or radio waves.	The Toy Train Tracks: The actual wooden tracks or wires that carry signals from place to place.

Section 8: Comprehensive Glossary of Computer Networking Terminology

- **Address Resolution Protocol (ARP):** A local link layer protocol used to dynamically map a known IP address to its corresponding physical MAC address.
- **Anycast:** A network routing technique where the same IP address is assigned to multiple global servers, routing client requests to the topologically nearest node.
- **Carrier-Grade NAT (CGNAT):** An IP sharing mechanism where an ISP maps thousands of private subscriber devices to a single shared public IP address.
- **Circuit Switching:** A communication method that reserves a dedicated physical connection between two endpoints for the entire duration of a session.
- **Cyclic Redundancy Check (CRC):** An error-detecting checksum stored in a packet trailer, calculated by summing payload bit values to detect corruption during transit.
- **Default Gateway:** The local interface of a Layer 3 router that host computers use to transmit data destined for outside networks.
- **Domain Name System (DNS):** A hierarchical distributed database system that translates human-friendly domain names into numerical IP addresses.
- **Dynamic Host Configuration Protocol (DHCP):** A network service that automatically assigns dynamic IP addresses, subnet masks, and network parameters to host devices.
- **Error-Correcting Code (ECC) RAM:** Specialized server memory that detects and corrects single-bit memory corruptions in real time to prevent system crashes.
- **Firewall:** A network security gateway device that monitors and filters incoming and outgoing traffic based on pre-defined security rules, IP addresses, and ports.
- **Hub:** An unintelligent Layer 1 central connecting device that blindly broadcasts incoming signals out to all connected ports.
- **Internet Message Access Protocol (IMAP):** An email retrieval protocol that synchronizes all email folders continuously across multiple devices while storing master copies on the server.
- **IP Address:** A 32-bit (IPv4) or 128-bit (IPv6) logical network layer address that uniquely identifies a device's location on an IP network.
- **Local Area Network (LAN):** A network connecting computers and devices within a limited physical geographical area, such as a home or office building.
- **Maximum Transmission Unit (MTU):** The largest protocol data unit (in bytes) that a layer or network medium can pass onward without fragmentation.
- **Media Access Control (MAC) Address:** A 48-bit hexadecimal physical hardware identifier permanently burned into a Network Interface Card (NIC) at the factory.
- **Network Interface Card (NIC):** A hardware component inside a host computer that provides the physical port or wireless antenna connection to a network.
- **Packet:** A standardized unit of data formatted with a header, payload, and trailer, transmitted independently across a packet-switched network.
- **Packet Switching:** A data transmission method where messages are divided into packets and routed dynamically across shared network paths.

• **Post Office Protocol 3 (POP3):** An email retrieval protocol that downloads inbox emails to a local client device and deletes them from the central server by default.

• **Redundant Array of Independent Disks (RAID):** A storage technology combining multiple physical hard drives to protect server data against single drive failures.

• **Router:** An intelligent Layer 3 gateway device that forwards data packets between separate networks using IP addresses and routing tables.

• **Simple Mail Transfer Protocol (SMTP):** A push-based application protocol used for transmitting outgoing emails between clients and servers, as well as between email servers.

• **Source Address Table (SAT):** A dynamic database maintained inside a Layer 2 switch mapping connected host MAC addresses to their physical switch ports.

• **Subnet Mask:** A 32-bit binary number used by IP hosts to distinguish the Network ID portion from the Host ID portion of an IPv4 address.

• **Switch:** An intelligent Layer 2 device that filters and forwards data frames directly to target host ports using MAC addresses.

• **Three-Way Handshake:** The 3-step synchronization algorithm (SYN, SYN-ACK, ACK) used by TCP to establish a connection session between two hosts.

• **Transmission Control Protocol (TCP):** A core transport-layer connection-oriented protocol that guarantees reliable, ordered, and error-checked byte stream delivery.

• **Virtual Private Network (VPN):** An encrypted connection tunnel built across a public network that hides a user's real IP address and secures transmitted data.

• **Wide Area Network (WAN):** A broad communications network extending across regional, national, or global boundaries (e.g., the Internet).